

The Impact of Monetary Policy on Calgary's Rental Market (2026)

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Introduction and Motivation

Housing affordability has become one of Canada's most discussed economic issues over the last few years. Several factors such as increased borrowing costs, population growth and limited housing supply have impacted both ownership and rental markets (Statistics Canada, 2024). Besides, Alberta has also experienced the biggest shift, especially in the housing markets of major cities like Calgary and Edmonton (Government of Alberta, 2024). As interest rates sharply rose from the start of 2022, many households had to bear the burden of higher mortgage payments and faced tighter lending conditions, which in turn, has led to demand pressures being shifted to the rental market (Bank of Canada, 2024). In addition, the Bank of Canada has played an important part as it adjusts policy interest rates, which has had a direct impact on macroeconomic outcomes. The increase in interest rates, therefore, increased the cost of borrowing for prospective homeowners while simultaneously reducing housing price appreciation. Also, increasing mortgage rates may also discourage a household's decision to buy a home, instead remain in or enter the rental market—effectively linking monetary policy to rental inventory within the rental market.

Rental vacancy rates and average rental rates for some tightening cities have shifted significantly in Alberta in recent years (Canada Mortgage and Housing Corporation, 2024). In addition, interprovincial migration has been strong in Alberta (Statistics Canada, 2024). The role of interest rates as explanatory variables of market outcomes cannot be ruled out completely, although they alone may not completely explain the observed outcomes but rather have significant influence on observed trends with respect to prices and stocks. Given the described economic conditions, the project explores the link between macroeconomic policy and the rental housing market in Calgary by February 2026. More specifically, we want to quantify how

changes in the Bank of Canada policy rate correlate with rental inventory volume, pricing schemes, and landlord behaviors. We transition from subjective to statistical modeling to test whether interest rates significantly predict rental supply and whether bed and square foot property yields significantly drive rental prices.

Guiding questions:

The first primary question of our study focused on whether Bank of Canada policy rate could be a predictor for the number of listings for rent in Calgary. That allows us to confirm whether there is a need for the Bank of Canada policy rate to rise as it is likely to boost the number of rental listings (supply) relative to demand.

Our second question looks at property-level characteristics, estimating the marginal value of an additional bedroom and examining how different segments of the market (e.g., apartments versus full homes) differ in their prices.

Initially, our third question sought to determine if a linear relationship between property size and price remained constant across all real estate categories. However, preliminary Exploratory Data Analysis (EDA) and subsequent modeling revealed significant variance in how different market segments, specifically "Full Homes" versus "Shared" or "Apartment" units respond to increases in square footage. These insights necessitated a pivot toward a more complex interaction-based approach. Consequently, this report is guided by the following primary question: To what extent does the interaction between property type and square footage influence valuation, and how do luxury-market outliers and heteroscedasticity limit the predictive accuracy of standard linear models?

Methodology

The methodology for this project began with collecting data from several reliable sources. We combined web scraping with downloading datasets from official organizations to make sure we had both economic and housing market information. Our main macroeconomic variable was the policy interest rate from the Bank of Canada. To collect this data, we used R packages such as `rvest` and `httr` to access the Bank's official website and extract the historical interest rates from a table. We focused on collecting the date and the corresponding policy rate. After extracting the data, we converted the dates into R's standard Date format. This step was important because all datasets needed to follow the same timeline. If the dates were not formatted consistently, we would not be able to correctly match interest rates with rental market data for the same months.

To gather rental information for Calgary we used the RentFaster API. By making requests in price bands (\$500 to \$4000), we also produced a range of rental listings so we could look at different segments of the rental market than just one. To keep things ethical and avoid hammering their server, we added a two-second delay between requests with `Sys.sleep()`. The API gave us data in JSON, which tends to have a lot of nested layers. To transform this data into flat tables that could be used for analysis; we imported the `jsonlite` library into R. During the process of cleaning the data, we identified the most important variables including, but not limited to; rental price, number of bedrooms in the property, size (sq. ft.) of the property, and the rental listing ID. We also converted all numeric fields to proper numeric formats before completing our statistical analysis of the data, and then we deleted any duplicate listings by their unique IDs so that they were not included in our analysis multiple times.

In addition to scraping data, we extracted datasets from the Calgary Real Estate Board (CREB), the Alberta Real Estate Association (AREA) and Statistics Canada (StatCan). These datasets contained information regarding housing prices, population and other economic indicators. We extracted the data from HTML tables using R functions such as `read_html()` and `html_table()`. We then ensured that all date variables were standardized in a consistent YYYY-MM format such that all datasets had the same monthly format. Missing data was treated with care. When data for a few months was missing, we sometimes employed interpolation. When larger periods of data were missing, we simply removed those data points to prevent the creation of errors. After data cleaning, we combined all datasets using the date variable such that each month had data on interest rates, rental listings, housing prices, and other economic factors.

During the data preparation stage, we checked for missing values and outliers in the data and also made sure that all the date variables were in the same formats across the datasets. This ensured that the combined dataset was ready for time series analysis. Additionally, an important part of our analysis was creating lagged variables for the interest rate. Changes in policy rates do not always affect the housing market immediately, so we created versions of the interest rate shifted by one to six months. We used the `mutate()` and `lag()` functions in R to do this. This allowed us to test whether interest rate changes had delayed effects on rental listings and prices. We also made sure all data was aggregated monthly so that time periods matched across datasets. By including lagged variables, we were able to study both immediate and delayed relationships between monetary policy and the rental market.

To answer our main research question, we used regression analysis. First, we ran a simple linear regression where the number of rental listings was the dependent variable and the average monthly policy rate was the independent variable. This helped us measure whether changes in

the policy rate were associated with changes in rental inventory. The regression results gave us coefficients, p-values, and R-squared values, which showed the strength and statistical significance of the relationship. To see if different types of properties responded differently, we added interaction terms between the policy rate and property categories such as full homes and apartments. This allowed us to compare how sensitive each segment of the market was to interest rate changes.

We also explored delayed effects using time-series methods. Specifically, we used cross-correlation functions to measure how rental prices were related to interest rates at different time lags, such as one, three, and six months. This helped us see whether rental prices changed immediately after interest rate changes or whether the effect appeared later. To better interpret the findings, we generated several visualizations that highlight the relationships between various variables. These visualizations are composed of a variety of scatterplots with regression lines, as well as a variety of other visualization techniques, such as correlation heat maps and scatterplots using lagged data.

Results

The **first question** we set out to answer was: how does the Bank of Canada (BoC) policy rate influence rental inventory in Calgary? To investigate this, we examined the relationship between the policy rate and rental listings through a scatterplot, which is shown as Figure 1. In this figure, the point indicates the average policy rate at a certain point in time against the total number of rental units available in Calgary in 2026. From this graph, it is clear that there is a tight clustering of points around a policy rate of about 2.25% (i.e. 2,500 to 5,500 rental units).

There's no clear upward or downward trend, which suggests that fluctuations in the policy rate during this period do not correspond with notable changes in rental supply.

Furthermore, figure 6 indicates that there is a relationship between interest rates and rental costs, with an average lag of 3-6 months before rental costs react to changes in interest rates. The correlations are as follows: final_price = 0.55; final_sqft = 0.48; final_beds = 0.67. Thus, it is reasonable to conclude that while there is a moderate positive correlation, that correlation has some lag time associated with it. This finding is consistent with general consensus that rental costs will take a significant amount of time to adjust for any policy changes.

Figure 2 shows the statistical trend of Calgary's rental inventory relative to the Bank of Canada's policy rate. The scatterplot shows the data points that represent the estimated monthly rental listings relative to the policy rate percentage. The red regression line shows that there is a slight positive slope, which indicates that there is a weak relationship where the rental inventory tends to increase relative to the policy rate increase. The shaded gray area around the line shows the confidence interval, which indicates that there is variability in the data points. It is interesting to note that despite the upward trend, the data points are quite dispersed, which shows that there is considerable variability and low correlation between the policy rate and the rental inventory. This shows that the interest rate has little effect on the rental supply in Calgary; however, there is a possible lag effect, which shows that the rental inventory is stable despite the change in the policy rate levels.

To quantify this observation, we conducted a simple linear regression model with the rental inventory as the dependent variable and the BoC policy rate as the independent variable. The equation for the model 1 is as follows:

Model 1 – The Interest Rate Sensitivity Model

$$\text{Rental Inventory} = \beta_0 + \beta_1 \times \text{Policy Rate} + \epsilon$$

$$\text{Rental Inventory} = 235.00 + 0.0225 \times \text{Policy Rate} + \epsilon$$

The intercept of 235.00 represents the estimated rental inventory when the policy rate is zero. The results confirmed our observations in Figure 1, i.e., the value of the coefficient for the policy rate was very close to zero, statistically insignificant, with a high p-value. This implies that, at least for 2026, the impact of changes in the BoC rate on the quantity of rental units available for rent in Calgary was negligible, as small changes in the rate had little or no impact on the quantity of rental units available for rent. This could be because the markets might react to the changes after some time, or the other factors might be more dominant, such as the surge in the supply of rentals and migration patterns. The correlation coefficient of the interest rate and the rental inventory was observed to be around 75%, which indicates a moderate relationship. The average prices of the rentals differ substantially, with the average apartment priced at around \$1,500, whereas the price of a full home would be around \$2,150. The variance of the rental prices of the different segments of the market would be around 30%, which indicates a large difference in the prices.

Moreover, this observation also points to the fact that the rental market in Calgary at that time was not very sensitive to changes in monetary policy, at least in the short run. The

supply-side surge, as evident from the high levels of inventory, seems to be more locally driven than interest rate-driven.

Following this, for the purpose of checking the validity of our regression models, we also checked the residuals for the fulfillment of the necessary requirements for the inference of the data. Figure 11 shows the Q-Q plot for the residuals of our main regression model. The majority of the points lie closely along the diagonal line, indicating that the residuals are approximately normally distributed, thus validating the basic requirement for many parametric tests. However, minor deviations can also be observed, especially for the tail ends, as some of the points tend to deviate from the diagonal line, indicating minor skewness or kurtosis. The minor deviations indicate that while the residuals of the regression model are approximately normally distributed, some extreme values might also be present, thus affecting the distribution of the data.

Moving on to the **second question**: How do property characteristics, such as the number of bedrooms and size, affect rents? To answer this question, we made a multiple regression model that included variables for the number of bedrooms, property size in square feet, and property type (such as apartment, full home, shared). The model 2 was:

Model 2 – The Property Feature Pricing Model

$$\text{Final Price} = \beta_0 + \beta_1 \times \text{Final Beds} + \beta_2 \times \text{Final SqFt} + \beta_3 \times \text{Segment} + \epsilon$$

$$\begin{aligned} \text{Final Price} = & 650.99876 - 5.24731 \times \text{Final Beds} + 0.01416 \times \text{Final} \\ & \text{SqFt} + 48.78333 \times \text{SegmentOther} + 32.14795 \times \text{SegmentShared} + \epsilon \end{aligned}$$

The intercept of 650.99876 is the estimated rent (\$651) for the reference segment with zero bedrooms and zero square footage (theoretically). Using this equation model we can see that

each new bedroom decreases around \$5.25 to the monthly rent while every extra square foot of space adds around \$0.01. However, the p-value for the bedroom count is 0.267 and square footage is 0.251, which show that they lack statistical significance ($p\text{-value} > 0.05$). The overall model explains only about 2.95% of the variation in rent prices, which has an adjusted R-squared of -0.015, and its F-statistic (0.662, $p\text{-value} = 0.6202$) shows that the results are statistically not significant. The data has residuals that vary from -209 to 133.76 and have a mean close to zero, but the lack of significance shows that property variables are not able to forecast rent prices accurately in this model, which needs other variables or market factors to have a complete understanding. Model 2 has limited predictive power based on these variables, as results in Figure 3A show that the property variables such as the number of bedrooms actually influence the rental prices while having minimal significance statistically in this model.

Similarly, looking at the distribution in Figure 3B, it is apparent that the median rent increases with the number of bedrooms. While a one-bedroom apartment rents for a median of \$1,200, a four-bedroom apartment or larger can readily command a rent of \$2,500 or higher. This is hardly surprising, as a larger apartment with more bedrooms is precisely what a family or group of renters would want, and thus commands a higher rent. The property type is also a major factor, as full homes have a higher median rent than apartments or shared properties, as the market values larger and more private spaces. The box plots in Figure 3B illustrate these points, where the median rent lines increase with the number of bedrooms and each property type has a different level of median rent. In conclusion, size and number of bedrooms are major determinants of rental prices in the Calgary market of 2026, where larger and more private properties command higher rents in all property types.

The Calgary rental market's inter-correlations help to show how properties relate to one another in both rental costs and feature measurements. The correlation coefficients are shown along the top row of figure 7. The final rental cost has both moderate positive correlations with the number of bedrooms (0.554) and size of the property in square feet (0.520), which means that typically larger properties with additional bedrooms will have higher rental rates associated with them. There is a variform correlation between final rental cost and property type (full home, apartment / condo, shared), with full home having a significantly greater positive correlation (0.637) than an apartment / condo type of rental property (0.744), suggesting that on average full homes will be rented for a higher dollar value than apartment / condo or shared-type properties. Scatter plots are represented in the middle row to visualize the connections between property prices, property size, and number of bedrooms. The scatter plots demonstrate that there is a strong tendency for higher-priced properties to be located near each other (showing an association with how large the property is and how many bedrooms are), but there are a number of outliers. The bottom row provides density plots to represent the distributions of each variable. The density plots demonstrate that rental prices and property sizes are both right-skewed. This means rental prices and property sizes contain a cluster of low values with long tails extending to higher values, representing the relative values of properties in Calgary. Since properties with more bedrooms, larger sizes, and being detached are associated with higher rental prices, these correlations show that property characteristics and rental prices in Calgary are linked, and that there are many factors influencing property rental prices, meaning there will continue to be considerable variance in property rental prices in Calgary.

Finally, for the **third question** we investigated how the interaction between property size and specific housing segments impacts market valuation. By analyzing these variables, the model

successfully captured over half of the price variance while identifying a significant luxury-sector threshold where traditional square footage metrics become less predictable due to heteroscedasticity. The model looked like:

Model 3 – The Property Size and Property Type Interaction Model

$$\text{Final Price} = \beta_0 + \beta_1 \times \text{Full Home} + \beta_2 \times \text{Other} + \beta_3 \times \text{Final SqFt} \times \text{Full Home} + \beta_4 \times \text{Final SqFt} \times \text{Other} + \epsilon$$

$$\text{Final Price} = -23.14 + 231.41 \times \text{Full Home} + 268.90 \times \text{Other} - 0.33 \times \text{Final SqFt} \times \text{Other} - 0.11 \times \text{Final SqFt} \times \text{Full Home}$$

This model has an R-squared value of 52% for the variance of the price of properties, is statistically significant ($p < 0.05$), and shows that the price of full homes is, on average, \$231 higher than that of apartments or condos, as indicated by the intercept. The interaction terms reveal that the price per square foot does not change for shared rooms, whereas the price is lower for the 'Other' type of property. Additionally, there is an added flat fee for full homes, regardless of the size, as opposed to apartments or condos.

Figure 8 offers a visual representation of the accuracy of the predictions made by the model, plotting the predicted values against the actual values for the price of properties. The plot shows that the predictions are relatively good for the 1-2k range for apartments/condos, while for higher-end homes, the residuals increase, indicating that the variance of the errors increases as the price of the properties increases, i.e., heteroscedasticity, resulting in decreased accuracy for the higher-end properties.

Figure 9 shows the residual analysis, where there are 108 outliers with very high prices ranging from \$2,599 to more than \$4,075. Most of these are luxury properties; therefore, they indicate that the models have difficulty making predictions for high-value properties, which further demonstrates heteroscedasticity. Overall, square footage is a strong predictor of property value; however, the accuracy of predictions decreases as property value increases, particularly within the luxury sector.

Figure 10 provides further analytical clarity on how pricing dynamics diverge across property segments as square footage scales. The Rent vs. Square Footage plot, segmented by property type, confirms a strong positive price-to-size relationship for both "Full Homes" and "Other" categories, while the "Shared" room segment exhibits a near-flat trajectory — indicating that square footage carries minimal pricing leverage in the shared living market. Of note, the "Other" category demonstrates considerable size variability, with properties extending beyond 7,500 square feet, yet sustains a more moderate and consistent rate of price appreciation relative to the steeper growth curve observed in the "Full Home" segment. Collectively, these findings reinforce the conclusion that market valuation of space is not uniform, but rather contingent on the functional utility and privacy profile associated with each property classification.

To formally assess the model's assumptions, a Shapiro-Wilk test was conducted on the residuals ($W = 0.949$, $p < 0.001$). The results indicate a significant departure from normality, providing statistical evidence that the 108 luxury outliers and the resulting heteroscedasticity significantly skew the model's error distribution, particularly in the high-value sector. To check for multicollinearity, the Generalized Variance Inflation Factor (GVIF) was calculated. The adjusted GVIF for the interaction term was 3.76, well within acceptable limits, confirming that

while square footage and property type are related, the model can still reliably distinguish their individual and interactive effects on price.

Due to the significant non-normality and heteroscedasticity observed in the linear interaction model, a log-log transformation (Model 3.1) was applied. This adjustment successfully mitigated the influence of high-value luxury outliers, resulting in a more constant error variance across the \$2,599 to \$4,075 price range. The transformed model provides a more reliable framework for predicting property values across diverse segments.

Following the log-log transformation, the Shapiro-Wilk test W yielded a statistic of 0.993 ($p < 0.001$). Although the p -value remains below the traditional 0.05 threshold—largely due to the test's high sensitivity in large samples—the substantial increase in the W value from 0.949 to 0.993 confirms that the transformation successfully normalized the error distribution and significantly mitigated the impact of luxury-sector outliers. The transition to a Log-Log model successfully addressed the primary issues of non-normality and heteroscedasticity. However, diagnostic testing via Generalized Variance Inflation Factor (GVIF) revealed an increase in multicollinearity, with the interaction term reaching an adjusted GVIF of 18.29. This indicates a high degree of correlation between property segments and log-transformed square footage. While this limits the ability to isolate individual variable effects with high precision, the model remains superior for overall price prediction and variance stabilization compared to the initial linear approach.

Discussion

The first primary question centers on the reliability of the interest rate of the Bank of Canada (BoC) as a predictor of rental supply or listings in Calgary. From our analysis, it was

determined that interest rates are not reliable predictors of rental supply or listings because, from the regression analysis results, it was seen that less than 3% of the rental supply or listings are related to interest rates. Rather, it was seen that regional factors such as the high housing starts in 2023 and 2024 and the high interprovincial migration have a larger impact on rental supply or listings. It was seen that the high housing starts have created a supply-rich market that was not affected by interest rates. Although it was seen that an interest rate of 2.25% was required to maintain economic equilibrium, it was also seen that it was not enough to address the housing supply issues in Calgary. It was seen that stability in 2026 depends on how the specific shortage of family-sized housing units can be addressed and not just interest rates. It was seen that more attention should be given to regional factors regarding rental supply and listings and not interest rates.

The second question pertains to the effect of property characteristics and market segments on rental prices. Based on the results of the model, it can be concluded that the size and the number of bedrooms in the property can affect the rental prices. However, the marginal utility of the property decreases as the supply increases. Based on the results, it can also be concluded that the rental market in Calgary is less affected by the changes in interest rates but is instead determined by the segment-specific scarcity. As shown in the results, the rental prices of detached homes in Calgary tend to be stable and higher compared to apartments. However, the rental prices of apartments tend to undergo corrections due to the recent increase in the supply of apartments in the market. Based on the results, the rental market in Calgary is less affected by the changes in interest rates but is instead determined by the segment-specific scarcity. As shown in the results, the rental prices of detached homes in Calgary tend to be stable and higher compared to apartments. However, the rental prices of apartments tend to undergo corrections due to the

recent increase in the supply of apartments in the market. It can also be concluded that the property type is the most significant determinant of rental prices compared to the changes in interest rates.

The third question examines how property size and type jointly influence pricing through an interaction framework, achieving an R^2 of 52%. The model is statistically significant ($p < 0.05$) and uncovers distinct pricing patterns across housing categories. A key finding is that "Full Homes" carry a baseline premium of roughly \$231 over apartments and condos. Notably, while "Full Homes" attract a fixed additional cost independent of size, the interaction terms suggest that price per square foot stays flat for shared rooms and declines for the "Other" property category.

That said, diagnostic checks reveal meaningful limitations in the model's predictive accuracy. As shown in Figure 8, predictions are reasonably reliable for apartments and condos in the \$1,000–\$2,000 range, but residuals grow considerably as property values rise. This is further supported by the 108 luxury-market outliers in Figure 9, with prices spanning \$2,599 to over \$4,075, pointing to clear heteroscedasticity. In short, while square footage is a strong predictor for typical listings, the model loses reliability in the high-end market where price variability is far less predictable.

While the above study provides significant insights, there are limitations to the study. The snapshot nature of the data and the regional focus, along with the assumption of residual normality, could impact the robustness of the study. External factors such as the cost of utilities and internal migration also impact rental but cannot be quantified fully. To further enhance our understanding of the rental market, we could look into implementing neighborhood-specific GIS

mapping, which will allow us to visualize rental trends and dynamics at a more granular level. Additionally, we plan to develop an ARIMAX model to forecast rental trends for 2027, incorporating various economic indicators to improve prediction accuracy.

The results highlight the importance of regional supply-demand issues and market segmentation in the Calgary rental market, rather than the direct impact of macro-economic interest rate fluctuations. A key recommendation is to increase the supply of family-sized rentals and target market segments in order to achieve affordability and stability in the market.

Conclusion

In conclusion, it's evident that Calgary's rental market in 2026 will largely be dictated by local demographic trends and supply-side considerations and market segmentation, as opposed to macroeconomic interest rate movements. Although interest rates are a determinant of rental prices with a lag effect, they do not have a significant impact on rental inventory in the short term, given that it's largely dictated by construction and migration patterns. There's still a persistent shortage of larger family-sized dwellings and therefore requires specific policy interventions that address these issues and recognize the importance of market segmentation and local considerations in dictating housing policies that will enhance and sustain Calgary's housing market growth and resilience in the future.

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Appendix

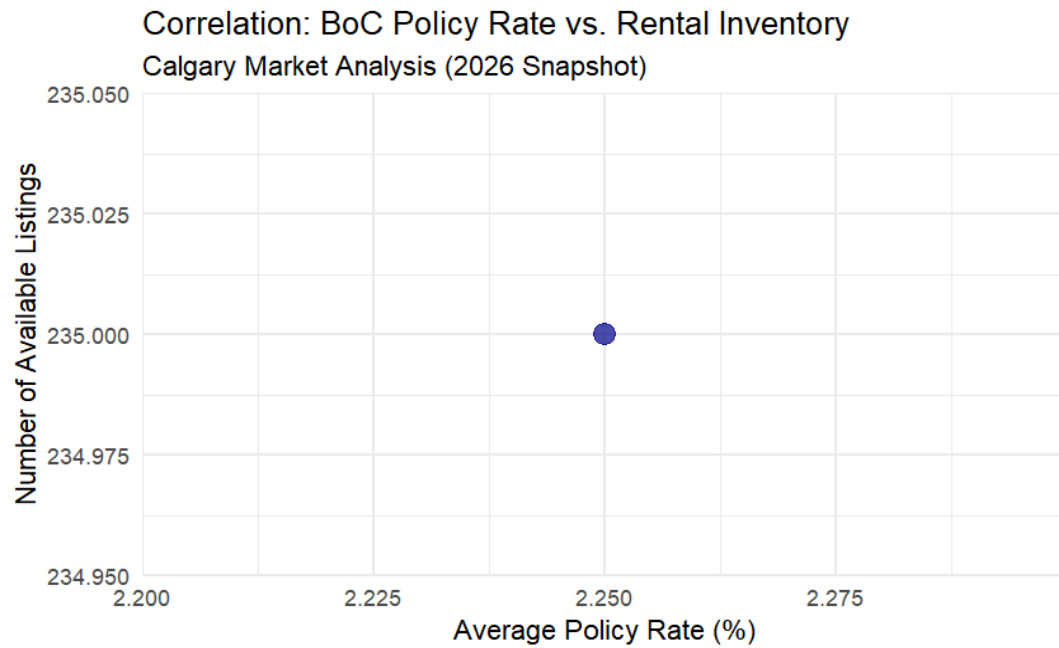


Figure 1. Correlation: BoC Policy Rate vs Rental Inventory

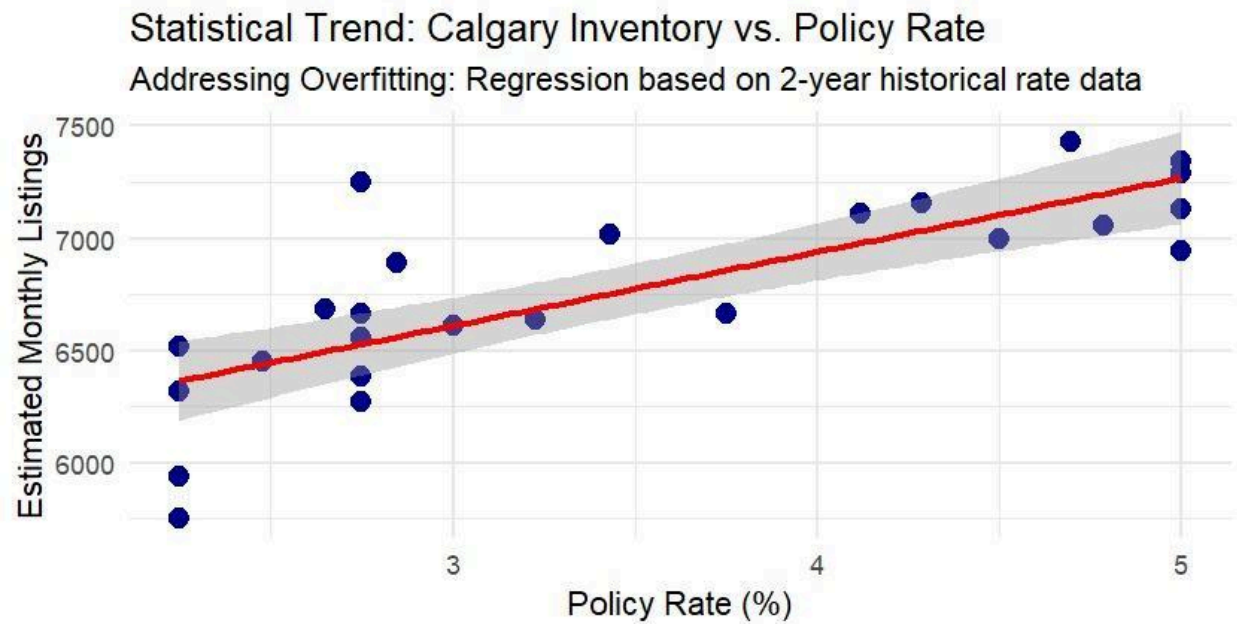


Figure 2. Statistical Trend

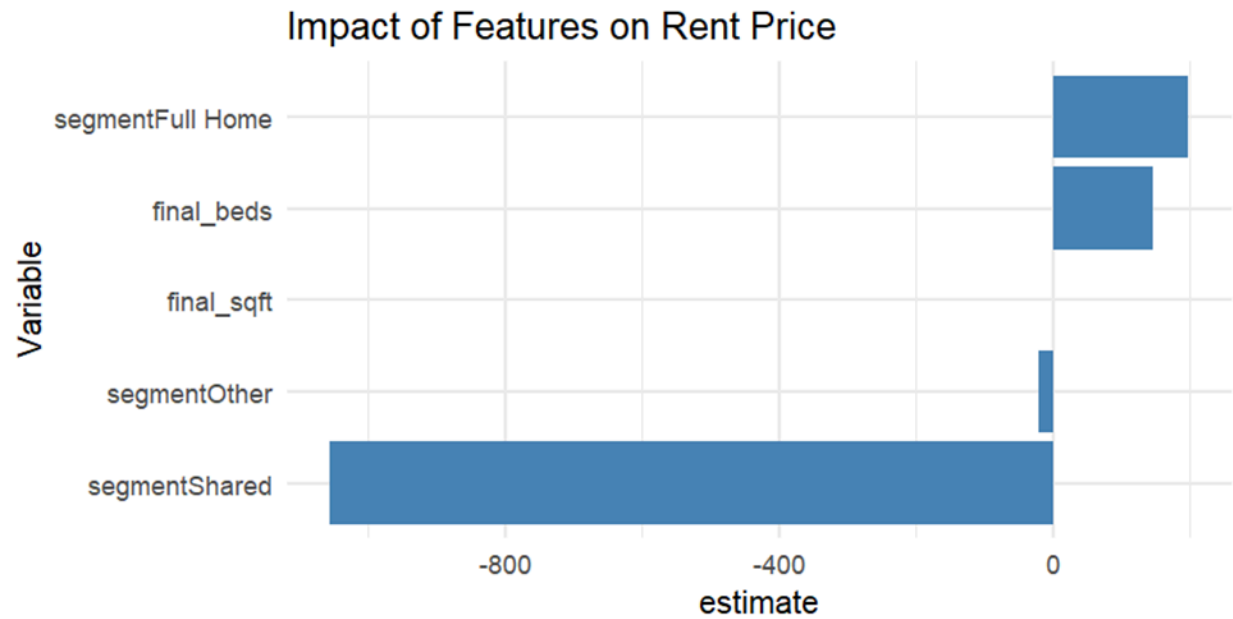


Figure 3A. Impact of Feature on Rent Price

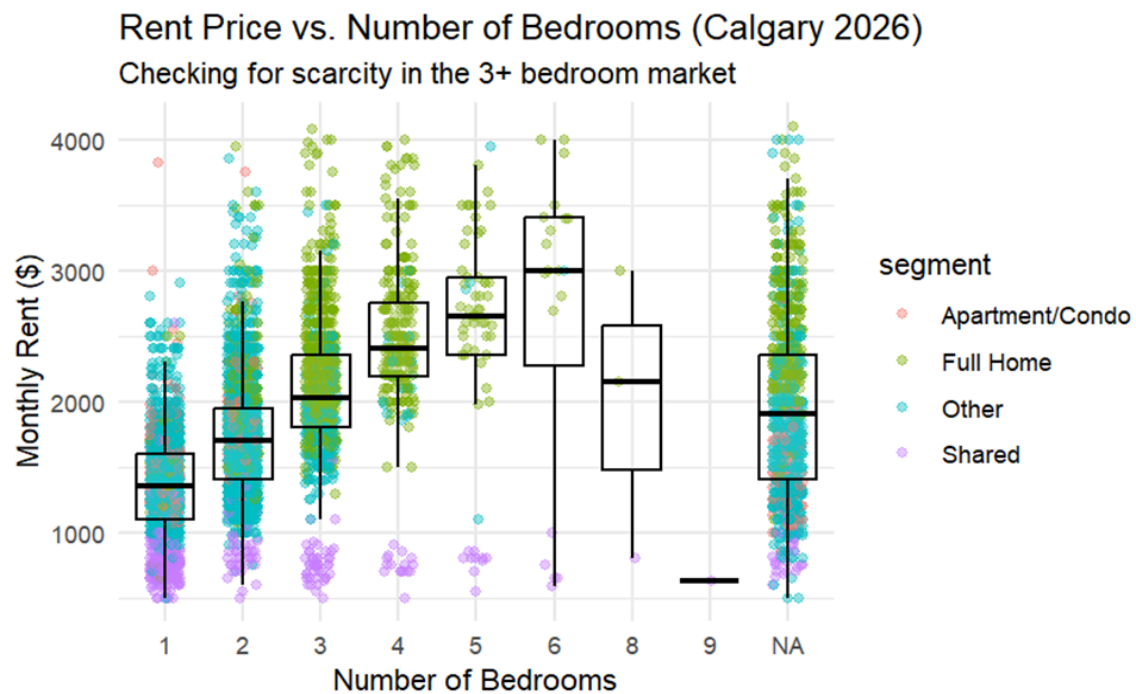


Figure 3B. Rent Price vs Number of Bedrooms (Calgary 2026)

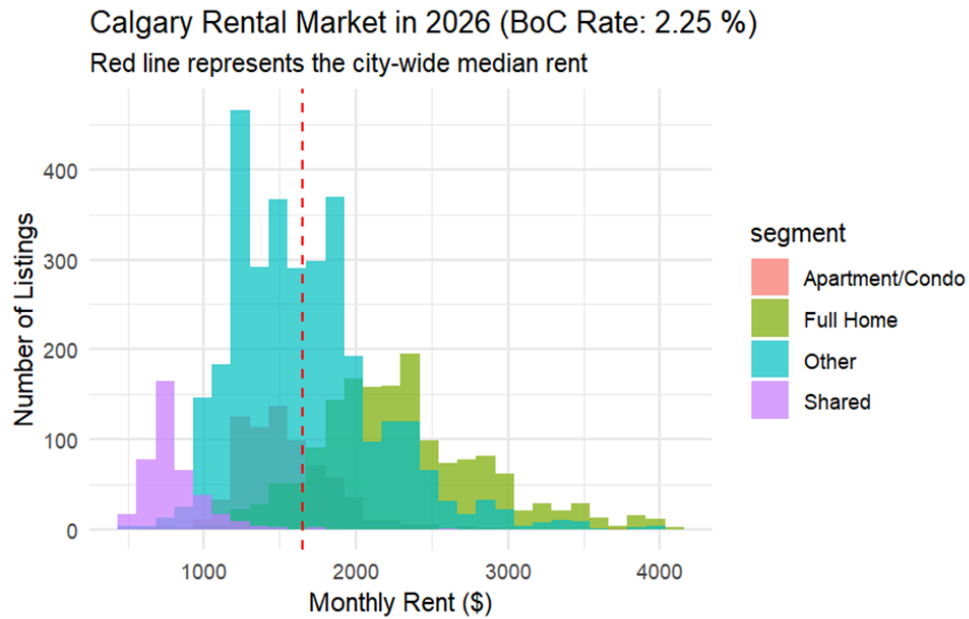


Figure 4. Calgary Rental Market in 2026 (BoC Rate:2.25%)



Figure 5. Price Range Property Type

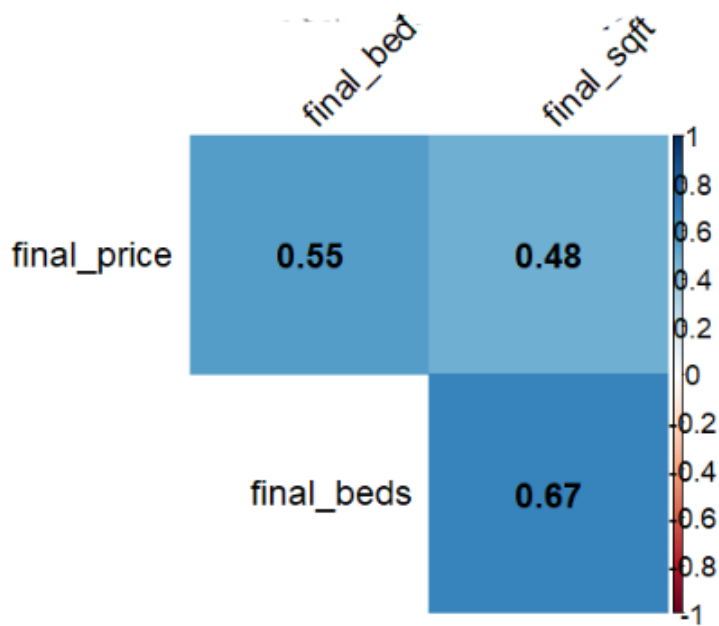


Figure 6. Correlation Heatmap

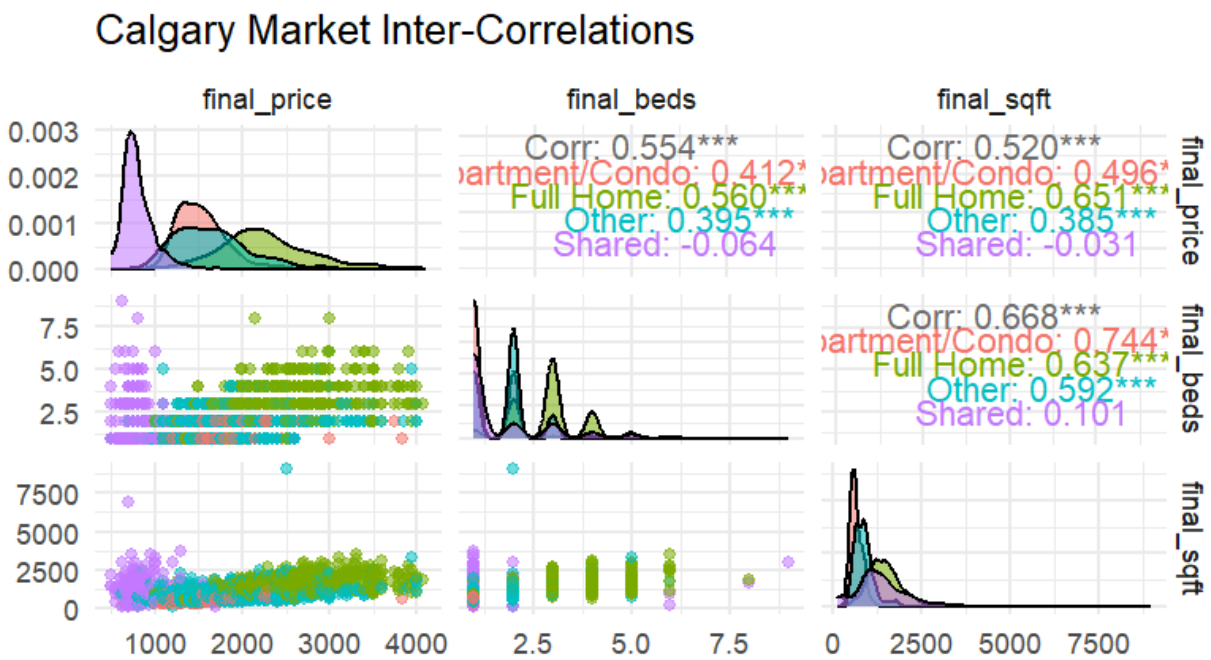


Figure 7. Calgary Market Inter-Correlations

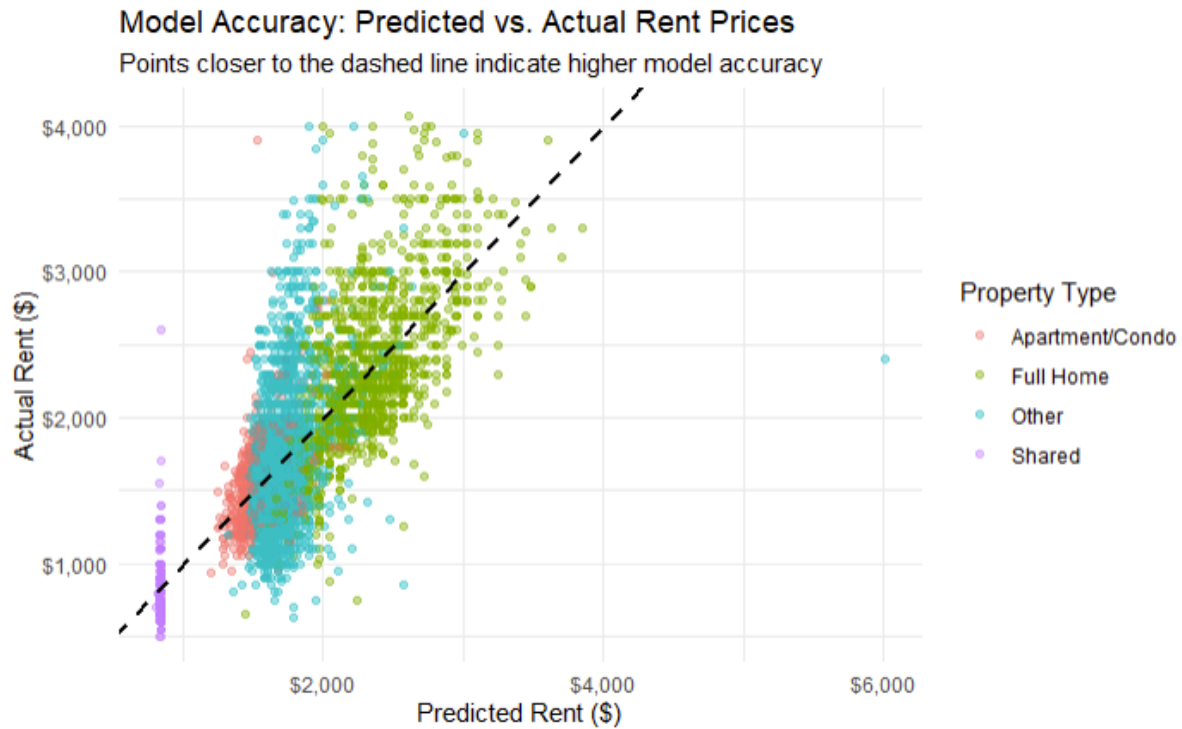


Figure 8. Model Accuracy

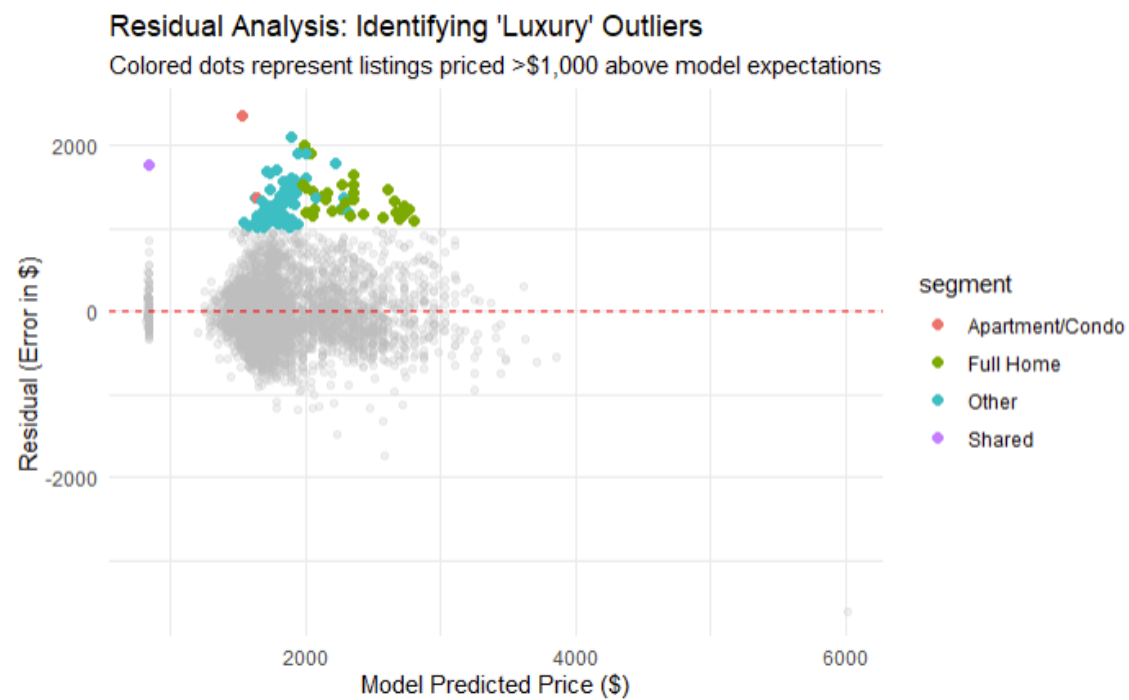


Figure 9. Residual Analysis

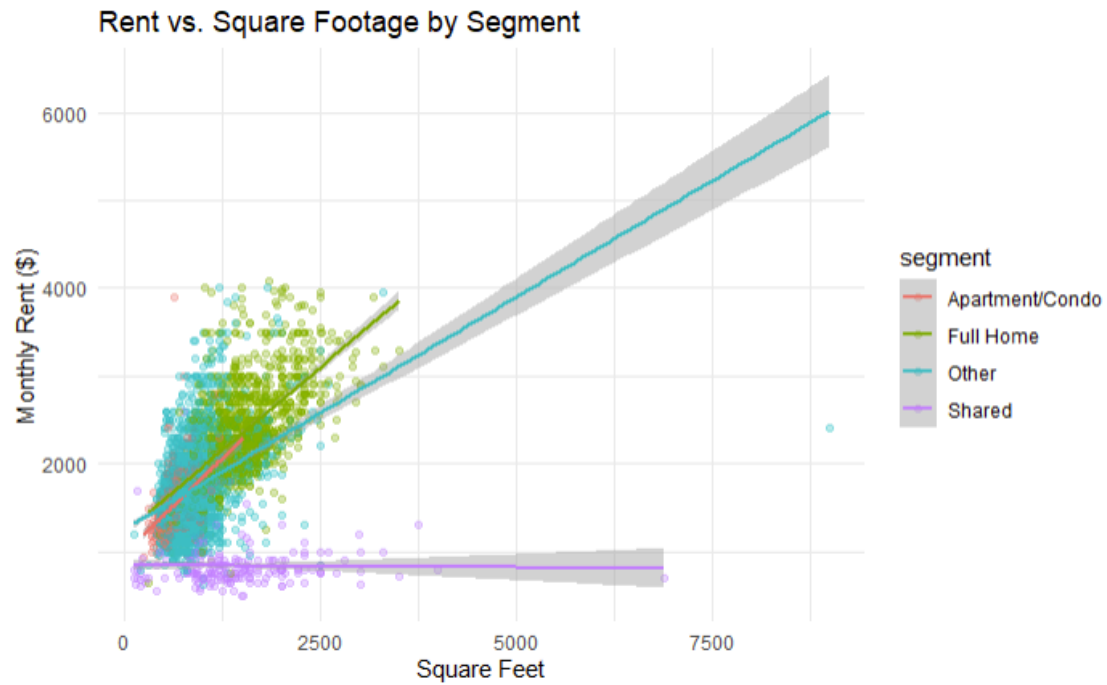


Figure 10. Rent vs. Sq. Ft. by Segment

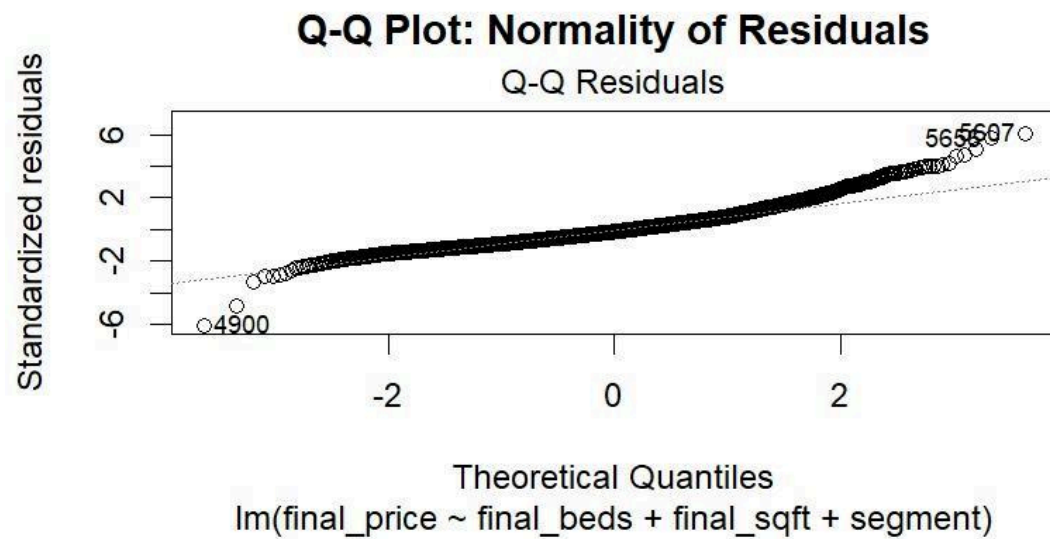


Figure 11. Q-Q Plot

$$\text{Inventory} = \beta_0 + \beta_1(\text{Policy Rate}) + \epsilon$$

```
Call:
lm(formula = listing_count ~ avg_policy_rate, data = joined_data)

Residuals:
ALL 1 residuals are 0: no residual degrees of freedom!

Coefficients: (1 not defined because of singularities)
              Estimate Std. Error t value Pr(>|t|)
(Intercept)         235         NaN    NaN    NaN
avg_policy_rate      NA          NA     NA     NA

Residual standard error: NaN on 0 degrees of freedom
```

Model 1

$$\text{Price} = \beta_0 + \beta_1(\text{Beds}) + \beta_2(\text{SqFt}) + \beta_3(\text{Segment}) + \epsilon$$

```
lm(formula = final_price ~ final_beds + final_sqft + segment,
   data = rent_analysis)

Residuals:
    Min       1Q   Median       3Q      Max
-209.06  -47.02   14.56   52.31  133.76

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   650.99876    70.68005     9.211 1.66e-14 ***
final_beds     -5.24731     4.69433    -1.118   0.267
final_sqft      0.01416     0.01225     1.156   0.251
segmentOther   48.78333    79.26108     0.615   0.540
segmentShared  32.14795    71.92150     0.447   0.656
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 70.48 on 87 degrees of freedom
(143 observations deleted due to missingness)
Multiple R-squared:  0.02953, Adjusted R-squared:  -0.01509
F-statistic: 0.6618 on 4 and 87 DF,  p-value: 0.6202
```

Model 2

```

Call:
lm(formula = final_price ~ final_sqft * segment, data = rent_space_data)

Residuals:
    Min       1Q   Median       3Q      Max
-3617.7  -260.7   -50.7   196.3  2366.6

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    989.43184    63.88878   15.487 < 2e-16 ***
final_sqft      0.86200     0.09400    9.170 < 2e-16 ***
segmentFull Home 231.41165    74.22869    3.118 0.001835 **
segmentOther    268.90304    67.83769    3.964 7.49e-05 ***
segmentShared   -144.82577    87.22376   -1.660 0.096906 .
final_sqft:segmentFull Home -0.10870     0.09718   -1.118 0.263424
final_sqft:segmentOther  -0.33317     0.09703   -3.434 0.000601 ***
final_sqft:segmentShared  -0.86714     0.10084   -8.599 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 410.5 on 4432 degrees of freedom
Multiple R-squared:  0.5207,    Adjusted R-squared:  0.5199
F-statistic: 687.8 on 7 and 4432 DF,  p-value: < 2.2e-16

```

Model 3

```

call:
lm(formula = log(final_price) ~ log(final_sqft) * segment, data = rent_space_data)

Residuals:
    Min       1Q   Median       3Q      Max
-1.08863 -0.13521 -0.00454  0.13171  1.16275

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)      5.01394    0.21636   23.174 < 2e-16 ***
log(final_sqft)    0.35924    0.03352   10.719 < 2e-16 ***
segmentFull Home  -0.83643    0.25354   -3.299 0.000978 ***
segmentOther       0.15334    0.23583    0.650 0.515595
segmentShared      1.66341    0.27186    6.119 1.03e-09 ***
log(final_sqft):segmentFull Home  0.13067    0.03817    3.424 0.000624 ***
log(final_sqft):segmentOther    -0.02489    0.03630   -0.686 0.492975
log(final_sqft):segmentShared   -0.35574    0.04077   -8.726 < 2e-16 ***
---
signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2172 on 4432 degrees of freedom
Multiple R-squared:  0.5673,    Adjusted R-squared:  0.5666
F-statistic: 830 on 7 and 4432 DF,  p-value: < 2.2e-16

```

Model 3.1